ABSTRACT

Research Project of the Bachelor degree in Information Technology

Emphasis in Network and Communication Systems

Adventist University of Central Africa (AUCA)

Title: Strategic Intelligence and Investigation Division System

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Name of Faculty Advisor: Mr.Ishimwe Prince

Date Completed: August, 2025

This project focuses on developing the Strategic Intelligence and Investigation Division System (SIIDS) for the Strategic Intelligence and Investigation Division in Rwanda Revenue Authority, the division responsible for overseeing intelligence gathering, surveillance, and investigations in tax fraud cases. The Division plays a crucial role in protecting national revenue by detecting fraud, enforcing compliance and coordinating investigation efforts. Traditionally, these processes have been largely manual, resulting in delays, data inaccuracies, security vulnerabilities, and inefficiencies in case tracking and resolution.

This project aims to address these challenges by providing a centralized, automated platform for case management, intelligence sharing, and investigation tracking. It enables intelligence officers, investigators, assistant commissioner and directors to securely access real-time data, collaborate across units, and streamline the approval and reporting workflow.

SIIDS is built with Spring Boot for backend processing, mySQL for database management, and a React.js frontend for an intuitive, role-based user interface. RESTful APIs enable integration with external government databases such as NIDA and RDB. By replacing manual workflows with a secure, integrated solution, SIIDS reduces administrative bottlenecks, enhances fraud detection capabilities, and improves operational efficiency in tax compliance and collecting national revenue.

DECLARATION

I, **Ntabana Nelson** with Registration Number **24562**, student at Adventist University of Central Africa in Faculty of Information Technology, Department of Software Engineering. I do hereby declare that this research project report is entirely the real reflection of my own original work and experience to the best of my knowledge. It has never been either partially or wholly presented in any university or any higher learning institution.

Signature…….

Date: …/……/……

APPROVAL PAGE

I, **Mr. Prince Ishimwe**, hereby certify that this project report, has been done under my supervision and submitted with my approval.

Signature: …….

Date: ……/……/……

DEDICATION

With great pleasure, I dedicate this Research Project

To my lovely parents,

To all my family members,

To all my friends and colleagues,

To my supervisor for his kind guidance.

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LIST OF SYMBOLS

AUCA: Adventist University of Central Africa

CSS: Cascading Style Sheet

DB: Database

DBMS: Database Management System

FK: Foreign Key

HTML: Hyper Text Markup Language

NIDA: National Identification Agency

RDB: Rwanda Development Board

RRA: Rwanda Revenue Authority

SIIDS: Strategic Intelligence and Investigation Division System

UML: Unified Modeling Language

ACKNOWLEDGEMENTS

I am especially grateful to my friends, classmates, and colleagues who supported me in countless ways. Your encouragement, teamwork, and kindness have been invaluable throughout this journey.

I would like to express my sincere thanks to my supervisor, Mr. Prince ISHIMWE, for his unwavering support, guidance, and constructive feedback throughout this academic journey. Your mentorship has played a crucial role in helping me complete my studies.

My heartfelt appreciation goes to the faculty and staff of the Faculty of Information Technology. Thank you for your dedication and for providing the knowledge, tools, and guidance that were essential to my growth and success.

I also extend my sincere gratitude to the administration and academic staff of the Adventist University of Central Africa (AUCA). Thank you for creating an environment that fosters both academic excellence and personal development. Special thanks to the lecturers in the Department of Networks and Communication Systems for their commitment to holistic education.

To my beloved parents, thank you for your unwavering love, prayers, and support throughout this journey. Your belief in me gave me the strength to persevere. I am equally grateful to my entire family for their encouragement, patience, and understanding during this challenging time.

Above all, I want to express my deep gratitude to the Almighty God, who made me who I am today. His grace, provision, and guidance sustained me throughout my studies and in every step of my journey.

Lastly, I would like to extend my appreciation to anyone else who directly or indirectly contributed to the completion of my studies.

May God bless you all abundantly.

NTABANA NELSON

CHAPTER 1

GENERAL INTRODUCTION

Introduction

The Strategic Intelligence and Investigation Division System (SIIDS) is designed to modernize intelligence, investigation, and surveillance processes within the Rwanda Revenue Authority (RRA). The digital transformation of government operations has become increasingly important in combating fraud, ensuring compliance, and improving operational efficiency. Tax fraud remains a significant challenge for revenue authorities worldwide, leading to substantial losses in government revenue and undermining economic stability.

Globally, many tax administrations have adopted advanced digital solutions to enhance their investigative capabilities. The integration of technology in intelligence gathering and fraud detection has proven to improve accuracy, speed, and security. However, RRA currently relies on manual processes, which limit the efficiency of intelligence operations and expose data to potential security threats.

Challenges such as limited accessibility to real-time data, inefficient case tracking, and inadequate collaboration between intelligence and investigation units highlight the need for an advanced system. SIIDS aims to bridge these gaps by automating intelligence and investigation workflows, ensuring secure access to data, and streamlining case management processes.

The implementation of SIIDS aligns with Rwanda's broader digital transformation goals, ensuring the RRA remains at the forefront of tax administration innovation. This project will enhance the ability to detect, investigate, and prosecute tax fraud, ultimately increasing compliance and revenue collection. (OECD)

Background of the study

Tax compliance is a critical pillar of sustainable national development, as governments rely on collected revenues to fund public services, infrastructure, and social programs. However, tax fraud and evasion continue to pose major threats to effective revenue collection, particularly in developing countries. In Rwanda, the Rwanda Revenue Authority (RRA) plays a central role in enforcing tax laws and ensuring compliance. Yet, the agency's current intelligence and investigation processes remain heavily reliant on manual workflows and limited data integration capabilities.

The Strategic Intelligence and Investigation Division System (SIIDS) of the RRA is responsible for detecting, analyzing, and investigating fraudulent tax activities. Due to the growing complexity and scale of tax-related crimes, there is an urgent need for technological solutions that can enhance the division's operational efficiency, accuracy, and responsiveness. Manual processes often lead to delays in case detection and resolution, inconsistencies in data handling, and difficulties in maintaining audit trails, all of which undermine the authority's ability to enforce compliance effectively.

In recent years, many tax administrations across the world have adopted digital systems that leverage automation, data analytics, and integration with national registries to streamline investigations and improve enforcement outcomes. Inspired by these global trends and aligned with Rwanda's broader digital transformation agenda, this study proposes the development of a platform the **Strategic Intelligence and Investigation Division System (SIIDS)**.

SIIDS is designed to modernize how intelligence is gathered, analyzed, and acted upon by RRA. It aims to automate case tracking, facilitate real-time data access from external sources like NIDA, RDB, and RLMUA, and enhance coordination between intelligence, surveillance, and investigation units. By introducing a secure, scalable, and intelligent system, the RRA can significantly improve its capacity to detect tax fraud, manage cases more efficiently, and ultimately foster greater public trust in the integrity of Rwanda's tax administration.

Statement of the Problem

The current tax intelligence and investigation system at RRA lacks a centralized and automated platform to manage intelligence gathering, investigation tracking, and case resolution. This deficiency results in inefficiencies, data inconsistencies, and security vulnerabilities, hindering the agency's ability to fight tax fraud effectively.

Manual processes contribute to delays in detecting fraudulent activities, difficulty in tracking cases, and limited access to real-time intelligence. The lack of integration with external databases such as NIDA and RDB further complicates investigations, making it challenging to validate taxpayer information promptly.

The impact of these inefficiencies is significant, leading to uncollected taxes, legal challenges, and diminished public trust in the tax administration system. Without a secure, automated, and integrated solution, RRA faces increasing difficulty in enforcing tax compliance and reducing fraudulent activities.

Choice and Motivation in the Study

The choice to focus on developing the Strategic Intelligence and Investigation Division System (SIIDS) stems from a critical need within the Rwanda Revenue Authority (RRA) to modernize and strengthen its tax investigation and fraud detection capabilities. Tax fraud is one of the leading causes of revenue loss in developing countries, and its detection requires timely access to accurate data, streamlined workflows, and secure collaboration across departments. Yet, the current manual and semi-automated systems at RRA are prone to inefficiencies, delays, and human error.

The motivation behind this study is rooted in the growing importance of digital transformation in public administration. Countries around the world are increasingly adopting intelligent systems to combat financial crimes, improve operational transparency, and increase efficiency in government services. Rwanda, through its Vision 2050 and National Digital Transformation Policy, emphasizes the adoption of ICT-driven solutions to enhance public sector performance. This project directly aligns with that national vision.

From an academic perspective, the study provides an opportunity to apply knowledge of software engineering, data integration, secure system design, and process automation to a real-world

challenge. Personally, the project is a chance to contribute meaningfully to Rwanda's digital governance efforts while advancing skills in system architecture, backend development, and investigative process automation.

Objectives of the Study

General Objective

- The primary aim to develop Strategic Intelligence and Investigation Division System (SIIDS) is to modernize and centralize intelligence gathering, investigation management, and enforcement processes at the Rwanda Revenue Authority.

Specific Objectives

- To enhance operational efficiency by automating manual processes including case registration, tracking, assignment, and status updates throughout the investigation workflow.
- To improve data accuracy and accessibility by integrating SIIDS with external databases (NIDA, RDB, RLMUA) for real-time taxpayer verification and comprehensive fraud detection.
- To strengthen system security and accountability by implementing role-based access controls, comprehensive audit trails, and secure data handling mechanisms to protect sensitive investigation information.
- To develop advanced fraud detection capabilities by incorporating data analytics tools and automated alert systems to identify suspicious tax activities and patterns more effectively.
- To optimize decision-making processes by providing real-time dashboards, automated reports, and analytical insights to support investigators and management in enforcement actions.
- To facilitate seamless communication and collaboration by implementing automated notification systems and centralized information sharing between intelligence, investigation, and enforcement units.
- To ensure system scalability and maintainability by designing a robust architecture capable of handling increasing data volumes and adapting to future technological and organizational requirements.

Scope of the Project

This section outlines the boundaries and extent of the Strategic Intelligence and Investigation Division System (SIIDS) project. It defines what functionalities and components are included in the system, as well as what falls outside the project's scope. Development of a web-based SIIDS platform.

- Integration with RRA's existing software and external databases.
- Automation of case tracking, intelligence gathering, and report generation.
- Implementation of role-based access control and security features.
- Training of RRA personnel on system usage and best practices.
- Ongoing system maintenance and support.
- Development of hardware infrastructure (existing infrastructure will be utilized).
- Broader tax administration functions outside of intelligence and investigation. *(OECD, 2021)*
- Manual data entry beyond initial system migration.

By implementing SIIDS, RRA will enhance its capacity to detect, investigate, and prosecute tax fraud, ensuring a more efficient and secure tax administration process. *(OECD, 2021)*

Methodology and Techniques

The development of the Strategic Intelligence and Investigation Division System (SIIDS) follows a structured methodology aimed at ensuring that the final solution is functional, user-centric, secure, and scalable. The approach combines established software engineering practices with real-world system development tools to meet the requirements of the Rwanda Revenue Authority (RRA).

Documentation

For this study, I examined various internal documents, procedures related to case reporting, and investigation at the Rwanda Revenue Authority (RRA). These included procedural manuals for handling tax fraud cases, sample case reports, and investigation handoff sheets. The review revealed several key issues: delays in manual case tracking, fragmented information across

departments, and poor visibility into case progress. These findings helped shape the direction of the SIIDS system to specifically automate case flow, enhance traceability, and integrate cross-departmental operations.

Interview

To gather deeper insights, I conducted informal interviews with RRA personnel involved in tax intelligence and investigation. These included Intelligence Officers, a system administrator, and an official from the Legal and Enforcement Unit. The aim was to understand the real-world challenges faced during case processing and fraud detection using the existing manual and semi-automated methods.

Key insights from interviews:

Officers mentioned that **case tracking is often delayed** because reports must be physically reviewed and approved step-by-step, sometimes across different offices.

There were repeated concerns about the **inability to access taxpayer data instantly**, especially when external verification (e.g., NIDA or RDB) was required.

Users emphasized the need for a system that could **send automatic alerts and allow real-time collaboration**, reducing reliance on calls or email follow-ups.

These interviews were instrumental in ensuring the SIIDS platform addresses the pain points experienced by end-users and aligns with their workflow expectations.

Sample Interview Highlights

Interview with an RRA Intelligence Officer

Q1: What is the most frustrating part of managing tax fraud cases under the current system?

A: "*The manual back-and-forth. You have to wait for files to be signed, reviewed, or corrected several times. Sometimes you even lose track of where the file is.*"

Q2: How would technology help improve the process?

A*:"* *If we could see the case progress, get instant alerts, and access taxpayer data right from the system, that would change everything. It would make the work faster and less prone to mistakes."*

Expected Results

The implementation of the Strategic Intelligence and Investigation Division System (SIIDS) is expected to produce a wide range of operational, technical, and organizational benefits for the Rwanda Revenue Authority (RRA). These results will directly address the limitations of the current manual processes and support more effective tax fraud detection and case resolution:

Improved Case Management Efficiency: SIIDS will automate the reporting, assignment, and tracking of investigation cases, reducing delays caused by manual processing.

Enhanced Fraud Detection Capabilities: By integrating external data sources and using analytical tools, SIIDS will enable more accurate and timely identification of fraudulent behavior.

Strengthened Accountability and Auditability: With full audit trail logging and role-based access controls, SIIDS will offer higher levels of security and traceability.

Better Informed Decision-Making: SIIDS will generate real-time dashboards and analytical reports, helping managers and decision-makers act based on data-driven insights

Organization of the Report

This study is organized into five chapters:

Chapter 1 General Introduction, providing an overview of the study, including the introduction, background, problem statement, motivation, objectives, scope, methodology, and expected results.

Chapter 2 Analysis of the Existing System, describing and analyzing the current processes, identifying issues, and presenting solutions.

Chapter 3 Requirements Analysis and Design of the New System, detailing the conceptual design and proposed solutions using UML methodology.

Chapter 4 Technical Realization of the Application, explaining the development process, technologies used, testing, and deployment.

Chapter 5 Conclusion and Recommendations, summarizing the study and offering suggestions for future development.

CHAPTER 2

ANALYSIS OF CURRENT SYSTEM

Introduction

The Rwanda Revenue Authority (RRA) currently relies on a predominantly manual and partially digitized process for handling tax intelligence, investigations, and enforcement. While functional at a basic level, this system faces significant limitations in terms of efficiency, data accuracy, security, and case traceability. The existing workflow involves physical documentation, face-to-face approvals, and inconsistent data sharing between departments, which hampers the agency's ability to respond swiftly and effectively to tax fraud cases.

This section analyzes the current system used within the Strategic Intelligence and Investigation Division (SIID) by identifying its core components, workflow structure, and operational weaknesses. The goal is to highlight the system's shortcomings that directly contribute to delays, data redundancy, and limited investigative capacity. By understanding how the current system operates and where it fails the need for a modern, centralized, and automated solution becomes clear.

The findings from this analysis serve as a foundation for justifying the design and implementation of the proposed SIIDS platform, which aims to address these critical gaps and enhance RRA's overall investigative performance.

Description of the Current System

The current system used within the Strategic Intelligence and Investigation Division System (SIIDS) at the Rwanda Revenue Authority (RRA) is primarily **manual and paper-based**, with limited automation and minimal integration between departments or external data sources. It involves multiple departments working independently to handle tax intelligence, case investigations, and fraud enforcement, often resulting in communication delays, data redundancy, and inconsistent workflows.

Workflow Overview

1. **Case Reporting**: A case begins when a suspected instance of tax fraud is reported manually, either through paper submission or physical communication.
2. **Intelligence Review**: An Intelligence Officer receives the report, conducts initial analysis, and creates a physical or digital summary document.
3. **Approval Hierarchy**: The analysis is passed manually to a Director for review, and if accepted, forwarded to the Assistant Commissioner and eventually to the Investigation Unit.
4. **Investigation & Reporting**: The assigned investigator collects evidence, conducts field work, and submits a report to be reviewed again through the same chain of command.
5. **Case Resolution**: Upon final approval, the case may be escalated to enforcement or closed if no fraud is confirmed.

Description of the Current System Environment

Historical Background

The Rwanda Revenue Authority was established under Law No 15/97 of 8 November 1997 as a quasi-autonomous body charged with the task of assessing, collecting, and accounting for tax, customs and other specified revenues.

Over the years, RRA has made significant strides in digital transformation, introducing electronic tax filing (e-Tax), mobile payment systems, and electronic billing machines (EBMs) to enhance service delivery. (rra)

However, within the **Strategic Intelligence and Investigation Division**, processes have largely remained **manual or semi-automated**. Intelligence gathering, case logging, fraud investigations, and approvals are still managed using physical files, spreadsheets, email communication, and internal memos. This legacy system has created bottlenecks in case resolution, limited real-time access to data, and increased the risk of data loss or manipulation. As fraud schemes become more sophisticated, the existing system struggles to keep pace, thereby affecting RRA's ability to respond swiftly and decisively.

Vision

To become a world-class efficient and modern revenue agency, fully financing national needs.

Mission

Mobilize revenue for economic development through efficient and equitable services that promote business growth.

Analysis of the Current System

The current system used by the Strategic Intelligence and Investigation Division (SIID) at the Rwanda Revenue Authority (RRA) relies heavily on manual operations. Case initiation, intelligence analysis, approvals, and investigation tracking are conducted using paper forms, email communication, and basic spreadsheets. These disconnected and outdated practices lead to delayed investigations, data inconsistencies, and lack of oversight. An in-depth analysis of the existing system highlights the inefficiencies and risks that justify the development of a centralized, intelligent system like SIIDS.

Modeling the Current System

The existing workflow can be described in the following sequential steps:

1. **Case Reporting**
   A suspected case of tax fraud is reported to the SIID via paper-based forms or basic emails. No formal digital entry system exists.
2. **Intelligence Officer and Surveillance Review**
   The report is received by an Intelligence Officer who performs a manual review and prepares a written or typed summary of findings.
3. **Director of Intelligence Review**
   The summary is submitted to the Director, who reviews it and either approves it for further investigation or returns it for clarification.
4. **Assistant Commissioner Review**
   If approved by the Director, the case is escalated to the Assistant Commissioner for evaluation and authorization of formal investigation.

5. **Investigation Officer Assignment**

   Upon approval, the Director of Investigation assigns the case to an officer, who conducts the investigation using fieldwork, interviews, and document analysis.

6. **Report Submission and Decision**

   The Investigation Officer submits a final report, which is reviewed by senior officials. If the findings confirm fraud, the case is sent to enforcement; if not, it is closed or revised.

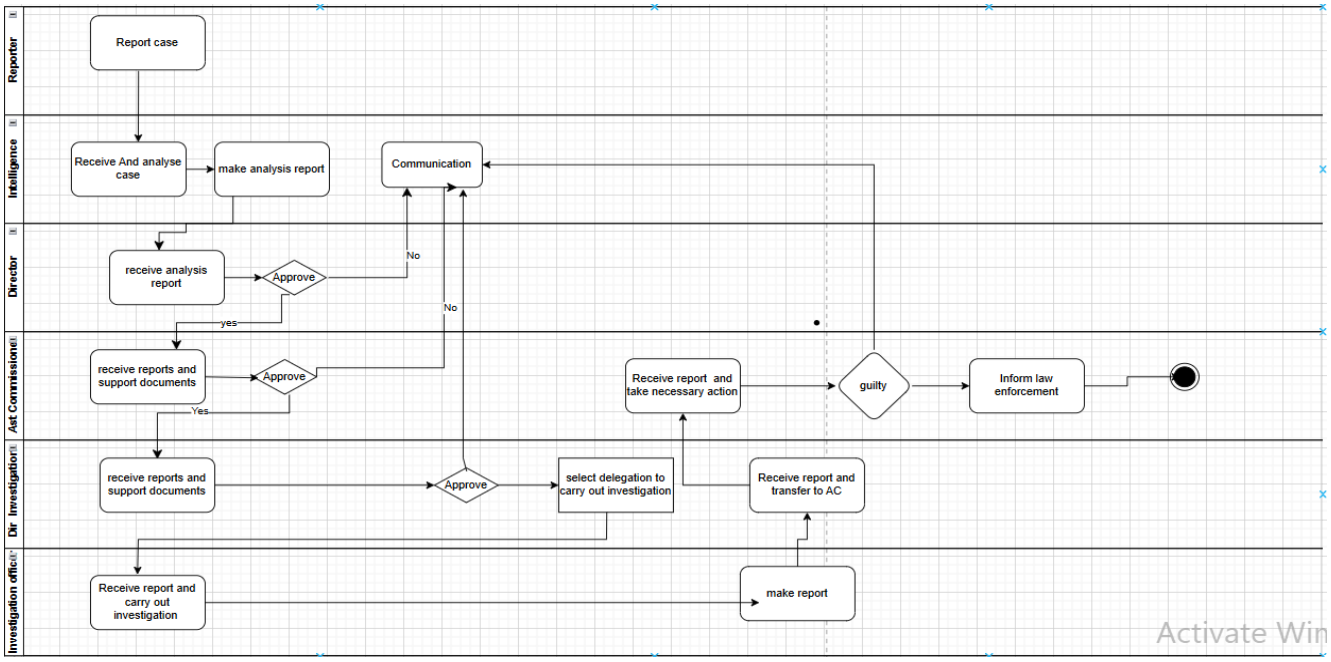


Figure 1: Current System

Problems of the Current System

Performance

- **Low Throughput:** The manual system processes a limited number of investigation cases per day due to paperwork and human dependencies. This restricts the volume of cases that can be completed within a given period, creating a backlog and slowing investigations.
- **High Response Time:** Case processing and approval times are significantly delayed as each step requires physical movement or manual communication between departments. This leads to longer wait times for case updates, decisions, and resolution.

Information

- **Storage Challenges:** Physical filing systems or scattered Excel files are used for data storage, posing risks of data loss, redundancy, and inaccessibility. There's no centralized digital repository to store, retrieve, or analyze case histories effectively.
- **Input Issues:** Data is often captured manually through handwritten or typed forms, which increases the chances of data entry errors and delays. There's no standardized format for input, leading to inconsistencies.
- **Output Limitations**: Reports and analysis results are manually generated and printed, making it difficult to quickly share findings or provide real-time insights. There is also a lack of automated alerts or dashboards for decision-makers.

Economic

- **High Operational Costs:** Printing, storing, and managing physical documents increases operational expenses.
- **Inefficient Resource Utilization:** More staff and time are required to manage and track cases manually, leading to increased labor costs.
- **Delayed Financial Decisions:** The slow processing of reports and approvals can lead to missed opportunities or delayed enforcement actions, affecting revenue collection.

Control

- **Weak Oversight & Tracking:** Tracking case progress and responsibilities is difficult, leading to accountability issues.
- **Risk of Fraud and Manipulation:** Manual handling increases the possibility of unauthorized alterations or misreporting.
- **Limited Audit Trail:** The system lacks an effective way to trace modifications or approvals, making investigations difficult in case of discrepancies.

Efficiency

- **Redundant Tasks:** Multiple approvals and case reviews require repeated manual verification, slowing down the process.

- **Difficulty in Collaboration:** Officers must rely on physical meetings or calls to discuss cases, leading to inefficiencies.
- **Delayed Decision-making:** Since reports require multiple approvals, cases take longer to reach resolution.

Service

- **Poor Customer Experience:** Individuals affected by enforcement actions face delays and frustrations due to slow case processing.
- **Lack of Transparency:** The status of cases is not easily accessible to relevant stakeholders, reducing public trust in the process.
- **Limited Scalability:** As case volumes increase, the system becomes more difficult to manage, leading to bottlenecks and increased delays.

Proposed Solutions

To address the inefficiencies, delays, and data management challenges found in the current manual system, this project proposes the development and implementation of a centralized and intelligent platform known as the **Strategic Intelligence and Investigation Division System (SIIDS)**. SIIDS is a web-based solution designed to automate and streamline the processes of intelligence gathering, fraud case tracking, inter-departmental communication, and enforcement actions within the Rwanda Revenue Authority (RRA)

SIIDS directly addresses these challenges by:

- **Automating manual processes**: Reduces processing time and human errors through digital case tracking and workflow management.
- **Enhancing real-time access to data**: Integrates external data sources for immediate verification and validation, solving the problem of data inaccessibility.
- **Improving security and accountability**: Implements strict access controls and audit trails, mitigating risks of fraud, manipulation, or unauthorized data access.
- **Supporting faster and smarter decision-making**: Provides dashboards, analytics, and alerts to investigators and decision-makers for proactive case handling.

System Requirements

The successful implementation of the Strategic Intelligence and Investigation Division System (SIIDS) requires a well-defined set of functional and non-functional requirements. These requirements ensure that the system is both technically sound and aligned with the operational needs of the Rwanda Revenue Authority (RRA).

Functional Requirements

These define the specific features and functions the SIIDS system must perform.

a) User Authentication and Access Control

REQ 1: The system shall allow users to log in using secure credentials.

|**REQ 2:** Role-based access shall restrict user permissions based on assigned roles (e.g., Intelligence Officer, Director, Investigator, Admin).

b) Case Management

REQ 3: Users shall be able to create, assign, update, and close investigation cases.

REQ 4: The system shall support attaching documents (e.g., reports, images, evidence) to each case.

REQ 5: Case history shall be tracked and viewable.

c) Workflow Automation

REQ 6: The system shall route cases to appropriate personnel based on predefined rules.

REQ 7: Automatic notifications and reminders shall be triggered for received cases.

d) Integration with External Databases

REQ 8: SIIDS shall connect retrieve data from NIDA, RDB, and RLMUA via secure APIs for identity, business, and property verification.

e) Real-Time Dashboards

REQ 9: Each user shall have access to a personalized dashboard displaying relevant metrics (e.g., open cases, notification alerts).

f) Reporting and Exporting

REQ 10: The system shall generate downloadable reports (PDF, Excel) for case summaries, investigation outcomes, and audit logs.

g) Audit Trail

REQ 11: Every system action (login, data edit, approval, etc.) shall be recorded with timestamps and user ID for audit purposes.

REQ 12: Logs Auditors shall have read-only access to all system activities and audit logs

Non-Functional Requirements

These define the quality attributes and constraints of the system.

a) Performance

REQ 1: The system shall be able to recover from a crash or failure within 9 minutes

REQ 2: Case queries and dashboard data shall load in under 3 seconds.

b) Security

REQ 3: The system shall ensure data security and privacy for all user data.

REQ 4: The system shall comply with relevant regulatory and legal standards.

d) Availability

REQ 5: The system shall maintain 99.5% uptime during business hours (Monday to Friday).

REQ 6: Scheduled backups shall occur daily.

e) Maintainability

REQ 7: The system shall provide regular backups to prevent data loss

f) Usability

REQ 8: The system shall provide a user-friendly interface that is easy to navigate.

g) Compatibility

REQ 9: The system shall be compatible with the latest versions of popular web browsers (Chrome, Firefox, Safari, Edge).

CHAPTER 3

REQUIREMENT ANALYSIS AND DESIGN OF THE NEW SYSTEM

Introduction

Designing an effective and secure investigative system for tax fraud detection requires a clear understanding of user needs, system limitations, and institutional goals. The **Strategic Intelligence and Investigation Division System (SIIDS)** is developed in response to the inefficiencies and gaps identified in the current manual processes at the Rwanda Revenue Authority (RRA). This section focuses on the analysis of system requirements and outlines the design framework used to build a scalable, secure, and intelligent platform.

Requirements analysis involves gathering, evaluating, and documenting the functional and non-functional expectations of end-users, system administrators, and organizational stakeholders. It ensures that the new system addresses real-world operational challenges, such as delayed case tracking, lack of integration with national databases, and poor visibility of investigation progress.

Following this, the design of SIIDS is guided by object-oriented principles and modular architecture to allow flexibility, future expansion, and maintainability. Unified Modeling Language (UML) diagrams such as use case, activity, and class diagrams are employed to visually communicate the system's structure and interactions.

The goal is to develop a system that not only meets technical standards but also empowers investigators, enhances transparency, and strengthens fraud detection capacity at the national level.

Unified Modeling Language (UML)

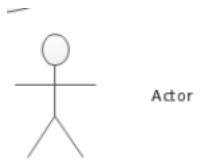
Unified Modeling Language (UML) is a standardized modeling language widely used in software engineering to visualize, specify, construct, and document the artifacts of a system. UML provides

a set of graphical notation techniques to create abstract models of the system, which are used to represent the system from different perspectives. In this section, several UML diagrams are utilized to depict the design of the new system. These diagrams include use-case diagrams, class diagrams, sequence diagrams, activity diagrams, database diagrams, and System Architecture Design. Each of these diagrams serves a specific purpose and provides unique insights into the system's architecture and interactions, ensuring that the design meets the specified requirements and addresses the shortcomings of the existing system, which will ultimately guide the effective implementation and deployment of the new system. The 18 comprehensive use of UML in this project underscores its importance in achieving a well-structured and functional system design. *(Sommerville, 2016)*

Use Case Diagram

A **Use Case Diagram** is a key part of system design that visually represents how **users (actors)** interact with the **system's features (use cases)**. It helps identify and organize the system's functional requirements based on user roles. For the Strategic Intelligence and Investigation Division System (SIIDS), the Use Case Diagram ensures that every type of user has access to the appropriate system functionalities.

Below are the symbols used in a use case diagram

Description	**Shape**
Actor: The use case entails detailing the sequence of actions that a system can undertake while interacting with external actors. It encompasses tasks that the system should perform in response to an actor's request and is visually depicted as follows:	
Use case: Each **use case** represents a distinct functionality or task that the system allows an actor to perform.	

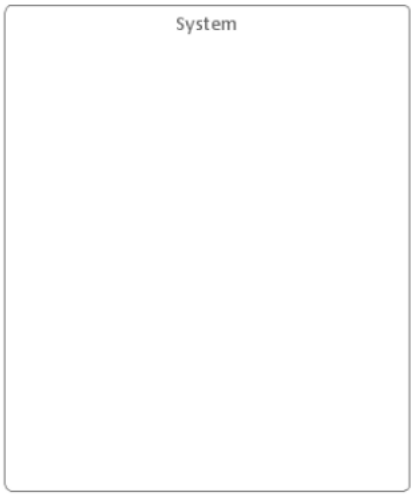
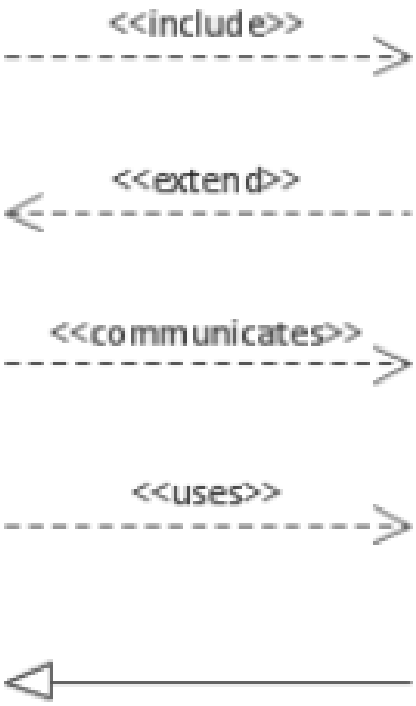
System boundary: This is a box drawn around the use case representing the edge or boundary of the system. It contains the name of the subject inside or on top.	
Relationship: Use cases are also connected to each other in different kinds of relationships. The relationship between two use cases basically models the dependencies between two use cases. An **association:** a relationship between two classifiers or describes the cause of the relationship and the rules that govern it A **generalization** relation is a relationship in which one model element (child) is based on another model element (parent). An **Include** relationship is a relationship in which one use case (base use case) contains the functionality of another use case (inclusion use case). an **extend** relationship to specify that one use case (extension) extends the behavior of another use case (base).	

Table 1:Symbols used in a use case diagram

Use Case Diagram of Strategic Intelligence and Investigation Division System(SIIDS)

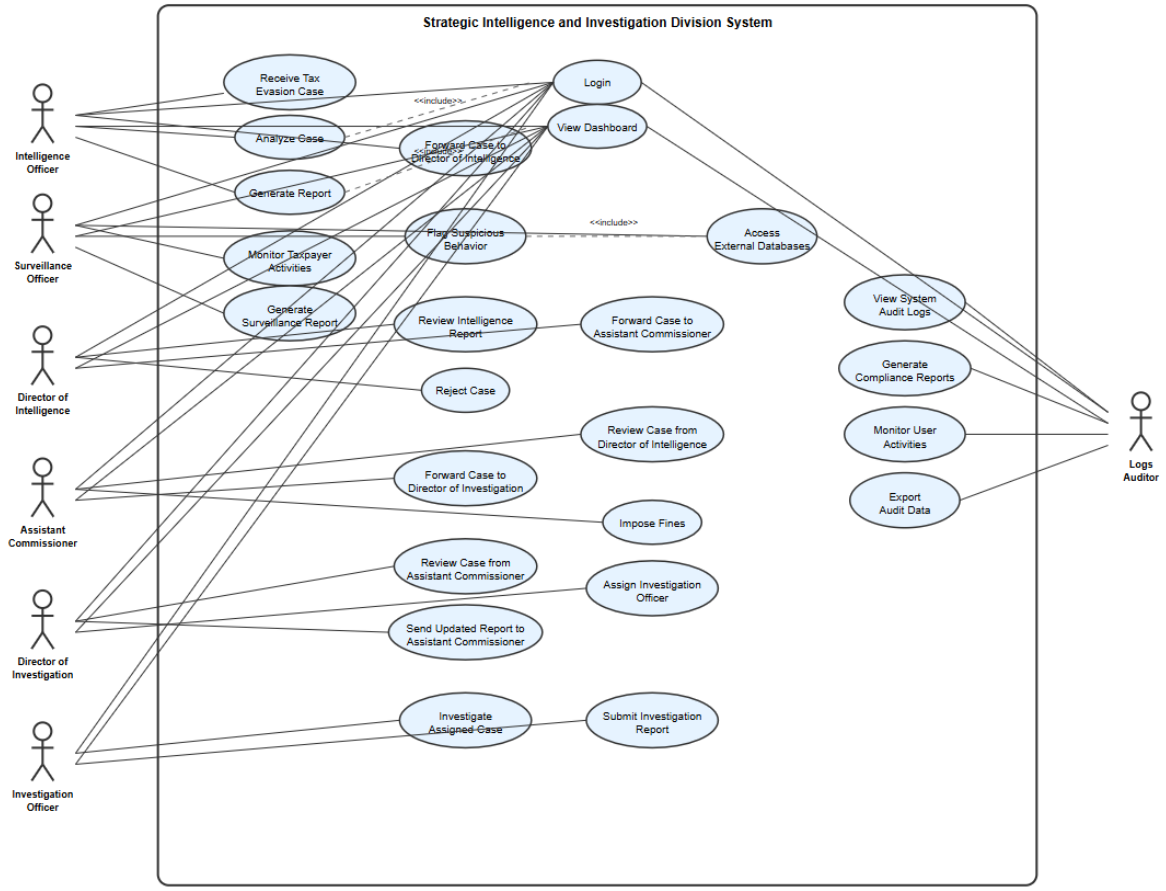


Figure 2: SIIDS use Case Diagram

Use Case Diagram Strategic Investigation and Intelligence Division System

Use-Case Description

A use case specification outlines the functionality and requirements for successful execution. The structure of each use case contains these essential elements:

Name: The identifier assigned to this specific use case

Actor: The actor or user who interacts with the system

Description: The primary purpose and functionality of this system operation

Pre-condition: Required system conditions that must exist before execution

Post-condition: Expected system conditions after successful completion

Normal flow: The sequential steps that occur during typical execution

Alternative flow: Alternative procedures activated when standard operations encounter issues

Use Case Name: Login
Actor: Intelligence Officer, Surveillance Officer, Director Intelligence, Assistant Commissioner, Director investigation, Investigation Officer
Description: Enables actors to access the system
Pre-condition: To be registered in the system
Post-condition: take action.
Normal Flow: 1. Accessing youndashboard
Alternative Flow None

Table 2: Login

Use Case Name: Receive Tax Evasion Case
Actor: Intelligence Officer
Description: Enables the Intelligence Officer to receive a new tax evasion case
Pre-condition: The Intelligence Officer is logged in.
Post-condition: The case is available in the system for further action.
Normal Flow:

2. The system notifies the Intelligence Officer of a new case. 3. The Intelligence Officer acknowledges receipt. 4. The system updates the case status to "Received."
Alternative Flow None

Table 3: Report case use case

Use Case Name: Analyze Case
Actor: Intelligence Officer, Director of Intelligence, Assistant Commissioner, Director of Investigation, Investigation Office
Description: Enables users to analyze case details and evidence.
Pre-condition: The case is registered and accessible.
Post-condition: Analysis results are recorded.
Normal Flow: 1. The user selects the case for analysis. 2. The system displays case details. 3. The user reviews and annotates findings. 4. The system saves the analysis. 5.
Alternative Flow None

Table 4: Analyze Case use case

Use Case Name: Generate Report
Actor: Intelligence Officer
Description: Allows the Intelligence Officer to generate a report based on case analysis.

Pre-condition: Case analysis is complete.
Post-condition: A report is created and stored.
Normal Flow: 1. The Intelligence Officer selects "Generate Report." 2. The system compiles analysis data into a report. 3. The user reviews and finalizes the report. 4. The system saves the report.
Alternative Flow None

Table 5: Generate Report case use case

Use Case Name: Forward Case to Director of Intelligence
Actor: Intelligence Officer
Description: Allows the Intelligence Officer to forward a case to the Director of Intelligence.
Pre-condition: A report is generated.
Post-condition: The case is sent to the Director of Intelligence.
Normal Flow: 1. The Intelligence Officer selects "Forward to DI." 2. The system attaches the report and sends a notification. 3. The Director of Intelligence receives the case.
Alternative Flow None

Table 6: Forward case use case

Use Case Name: Review Intelligence Report
Actor: Director of Intelligence
Description: Enables the Director of Intelligence to review the submitted report.
Pre-condition: The case is forwarded by the Intelligence Officer.
Post-condition: The report is marked as reviewed.
Normal Flow: 1. The Director of Intelligence opens the report. 2. The system displays the report details. 3. The Director approves or requests changes.
Alternative Flow 3.1: If changes are requested, the system returns the case to the Intelligence Officer.

Table 7: review Intelligence report case use case

Use Case Name: Forward Case to Assistant Commissioner
Actor: Director of Intelligence
Description: Allows the Director of Intelligence to escalate the case to the Assistant Commissioner.
Pre-condition: The report is approved.
Post-condition: The case is sent to the Assistant Commissioner.
Normal Flow: 1. The Director selects "Forward to AC." 2. The system notifies the Assistant Commissioner.

Alternative Flow • *None*

Table 8: Forward Case to Assistant Commissioner report case use case

Use Case Name: Reject Case
Actor: Director of Intelligence, Assistant Commissioner, Director of Investigation
Description: Allows authorized users to reject a case due to insufficient evidence.
Pre-condition: The case is under review.
Post-condition: The case is marked as rejected.
Normal Flow: 1. The user selects "Reject Case." 2. The system prompts for a rejection reason. 3. The user provides comments. 4. The system updates the case status.
Alternative Flow • *None*

Table 9: Reject case use case

Use Case Name: Review Case from Director of Intelligence
Actor: Assistant Commissioner
Description: Enables the Assistant Commissioner to review cases forwarded by the Director of Intelligence.
Pre-condition: The case is escalated by the Director of Intelligence.
Post-condition: The case is approved or redirected.

Normal Flow: 1. The Assistant Commissioner opens the case. 2. The system displays the report. 3. The Assistant Commissioner evaluates and takes action.
Alternative Flow • *None*

Table 10: Review Case from Director of Intelligence case use case

Use Case Name: Forward Case to Director of Investigation
Actor: Assistant Commissioner
Description: Allows the Assistant Commissioner to assign the case to the Director of Investigation.
Pre-condition: The case requires further investigation.
Post-condition: The case is sent to the Director of Investigation.
Normal Flow: 4. The Assistant Commissioner selects "Forward to Director of Investigation." 5. The system notifies the Director of Investigation
Alternative Flow • *None*

Table 11: Forward Case to Director of Investigation case use case

Use Case Name: Review Case from Assistant Commissioner
Actor: Director of Investigation

Description: Enables the Director of Investigation to review cases assigned by the Assistant Commissioner.
Pre-condition: The case is forwarded by the Assistant Commissioner.
Post-condition: The case is marked for investigation.
Normal Flow: 1. The Director of Investigation opens the case. 2. The system displays case details. 3. The Director assigns an Investigation Officer.
Alternative Flow • *None*

Table 12: Review Case from Director of Intelligence case use case

Use Case Name: Assign Investigation Officer
Actor: Director of Investigation
Description: Allows the Director of Investigation to assign an officer to investigate the case.
Pre-condition: The case is under the Director of Investigation's purview.
Post-condition: An Investigation Officer is assigned.
Normal Flow: 1. The Director selects an officer from the list. 2. The system notifies the assigned officer.
Alternative Flow 1.1: If no officers are available, the system alerts the Director.

Table 13: Assign Investigation Officer Case use case

Use Case Name: Send Updated Report to Assistant Commissioner
Actor: Director of Investigation
Description: Enables the Director of Investigation to send updated findings to the Assistant Commissioner.
Pre-condition: The investigation is complete.
Post-condition: The report is updated and sent.
Normal Flow: 1. The Director selects "Send Updated Report." 2. The system attaches the report and notifies the Assistant Commissioner.
Alternative Flow • *None*

Table 14: Send Updated Report to Assistant Commissioner Case use case

Use Case Name: Investigate Assigned Case
Actor: Investigation Officer
Description: Allows the Investigation Officer to conduct a detailed investigation.
Pre-condition: The officer is assigned to the case.
Post-condition: Findings are documented.
Normal Flow: 3. The officer accesses the assigned case. 4. The system displays case details. 5. The officer collects evidence and updates the case log.

Alternative Flow • *None*

Table 15: Investigate Assigned Case use case

Use Case Name: Submit Investigation Report
Actor: Investigation Officer
Description: Enables the Investigation Officer to submit the final investigation report..
Pre-condition: The investigation is complete.
Post-condition: The report is submitted for review.
Normal Flow: 6. The officer selects "Submit Report." 7. The system validates the report format. 8. The report is sent to the Director of Investigation.
Alternative Flow 2.1: If validation fails, the system highlights errors for correction.

Table 16: Submit Investigation Report case use case

Use Case Name: Impose Fines
Actor: Assistant Commissioner
Description: Enables the Director of Investigation to send updated findings to the Assistant Commissioner.
Pre-condition: The investigation confirms tax evasion
Post-condition: Fines are recorded and notified to the offender.

Normal Flow: 1. The Assistant Commissioner selects "Impose Fines." 2. The system calculates the fine amount. 3. The Assistant Commissioner approves and issues the fine.
Alternative Flow • *None*

Table 17: Impose Fines case use case

Use Case Name: View System Audit Logs
Actor: Logs Auditor
Description: Enables the Director of Investigation to send updated findings to the Assistant Commissioner.
Pre-condition:
Post-condition: Review comprehensive system activity logs for compliance.
Normal Flow: Access audit interface, filter logs, generate reports
Alternative Flow • *None*

Table 18: View System Audit Logs

Class Diagram

Introduction

Within UML modeling, a class diagram functions as a structural blueprint that illustrates system organization by displaying its component classes, along with their properties, functions, and the interconnections between different system elements. This type of visualization falls under the category of fixed architectural representations. (Cantu, 2018)

Tools to be used

Terms used and their definition	Symbol
An attribute: ❖ **Defines the characteristics that reflect an object's current status** ❖ **May be calculated using other existing properties, indicated by adding a forward slash at the start of the property name**	**Class Name** +attribute1: int -attribute2: float #attribute3: Circle
An operation: ❖ **Encompasses the behaviors and capabilities available to class instances** ❖ **Falls into 3 categories: initialization methods, data retrieval, and state modification procedures**	**Class Name** +op1 () -op2 () #op3()
An association: ❖ **Shows how different classes interact with each other or how a class relates to itself** ❖ **Can link two or more distinct classes together.**	
A generalization: ❖ **Demonstrates how one class serves as a specialized version of another class(parent-child relationships between classes)**	

An aggregation: ❖ **Indicates when elements of one class form part of another class's structure**	

Table 19: Class Diagram

Class Diagram of Strategic Intelligence and Investigation Division System(SIIDS)

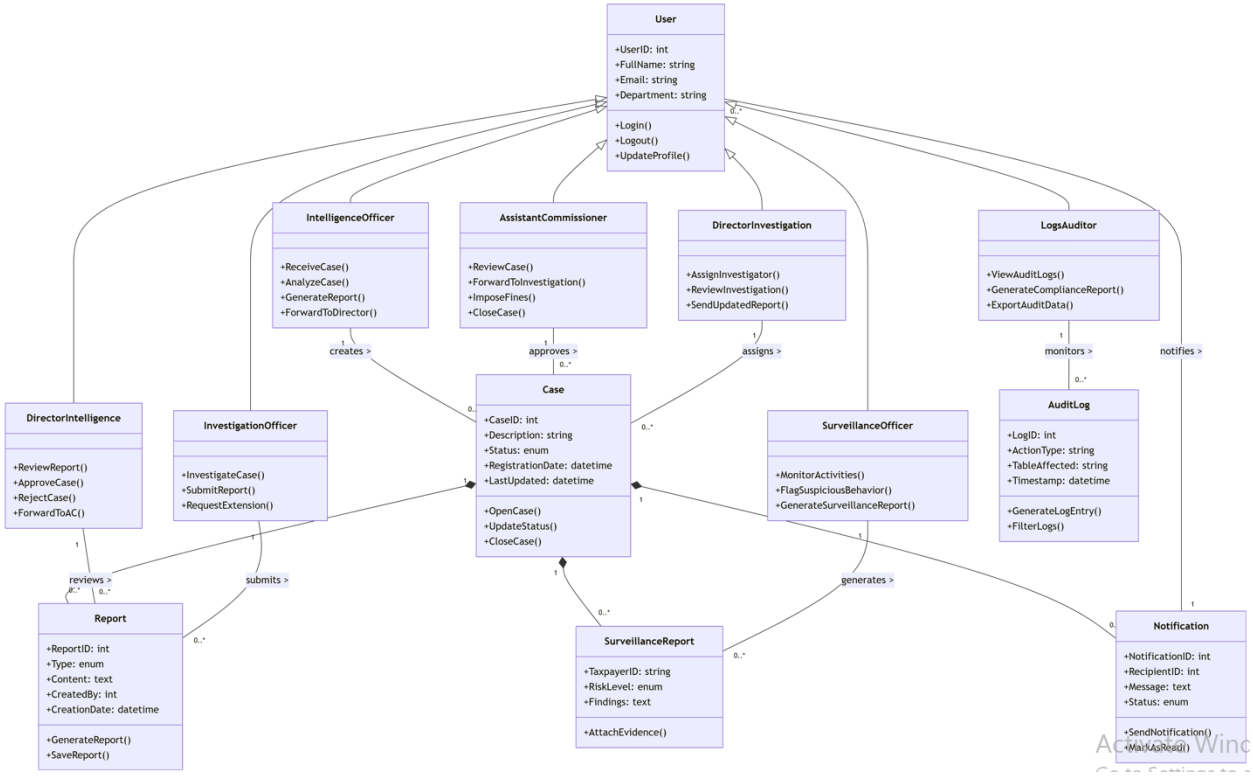


Figure 3: Class Diagram

Sequence Diagrams

Interaction flows are visualized through sequence diagrams, which detail the specific order and manner in which objects collaborate. Software engineers and business analysts utilize these visual representations to interpret system requirements and document operational processes. These diagrams also go by alternative names such as event flowcharts or scenario mapping diagrams. (Markham, 2018)

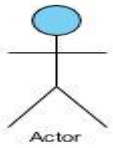
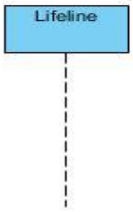
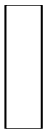
Terms used and their definition	**Symbol**
An actor: ✓ Represents an external individual or system that receives value from the system ✓ Engages in the process through message exchange ✓ Appears at the diagram's upper section	
An object lifeline: ✓ Takes part in sequences through message transmission and reception ✓ Positioned along the diagram's upper portion.	
An activation: ✓ Displayed as an elongated rectangular shape overlaid on the timeline ✓ Indicates the duration when an object is actively exchanging messages	
A message: ✓ Transfers data between system objects ✓ Function calls use solid arrows with action labels, while responses feature dashed arrows with return value descriptions.	: Message send Reply Message Self-Message

Table 20: Meaning of symbols used on sequence diagrams

Sequence Diagram of Strategic Intelligence and Investigation Division System(SIIDS)

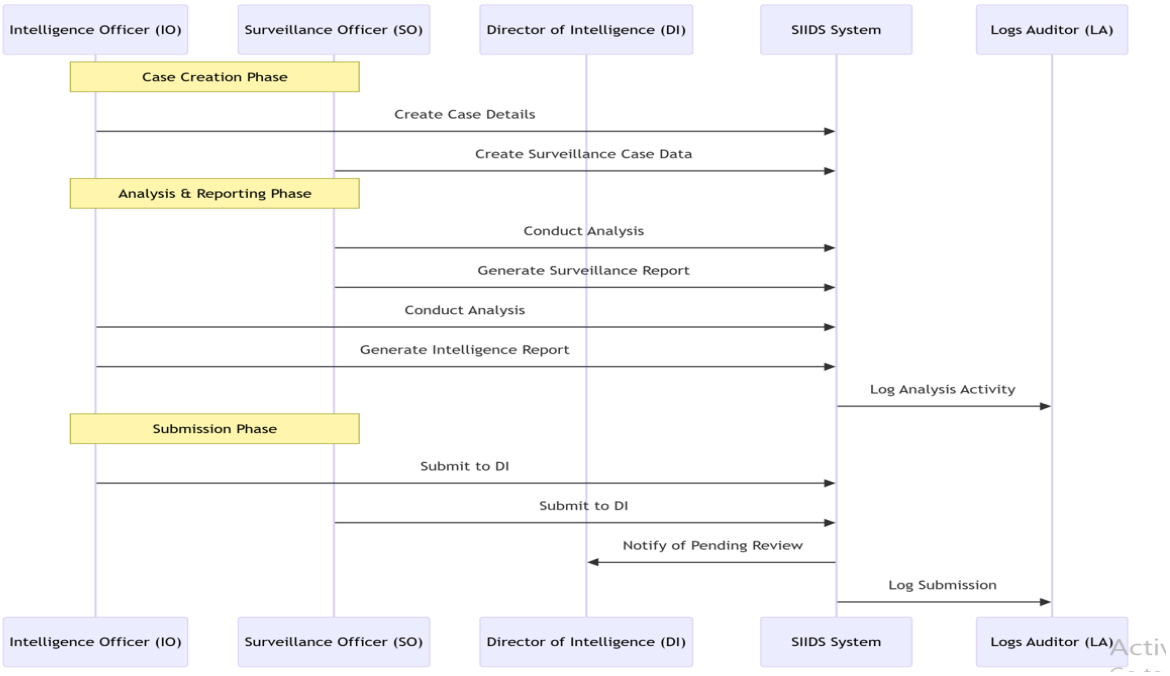


Figure 4: Case Initiation & Report Generation

Description:

- **Actors Involved**: Intelligence Officer, System
- **Process Flow**:

1. The **Intelligence Officer** initiates a new case by submitting details (e.g., taxpayer information, suspected fraud type).
2. The **System** validates the input, creates a case record, and assigns a unique CaseID.
3. The officer analyzes the case and requests a report.
4. The **System** compiles data, generates a report, and stores it in the database.
5. The officer reviews and finalizes the report.

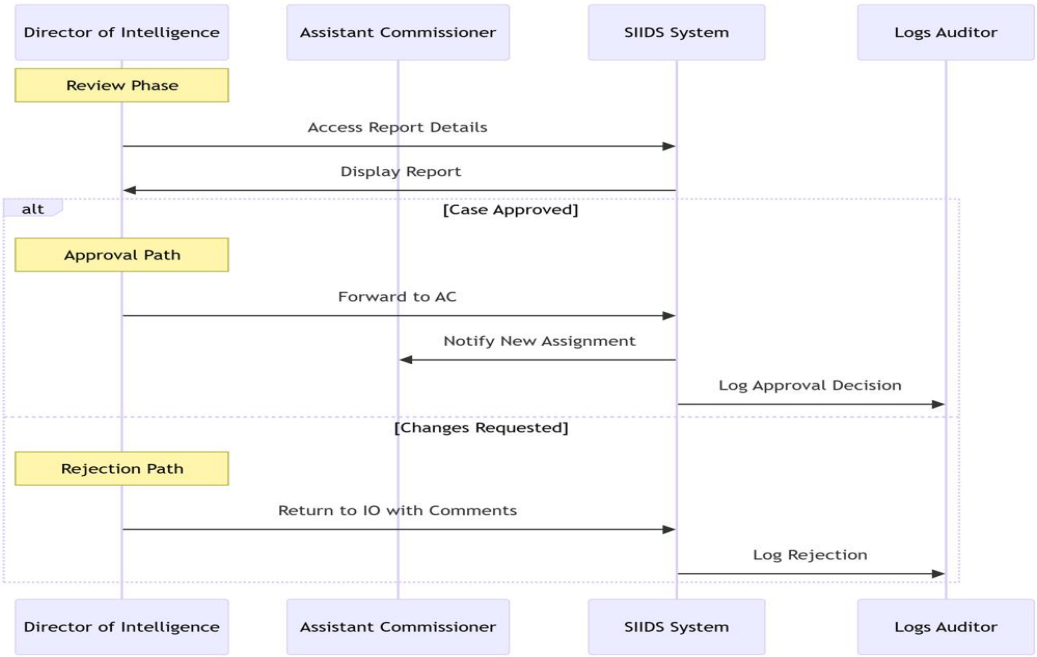


Figure 5: Director Review & Approval

Description

- **Actors Involved**: Director of Intelligence, System, Assistant Commissioner
- **Process Flow**:

1. The **Intelligence Officer** forwards the case to the **Director of Intelligence**.
2. The **System** notifies the director and displays the report.
3. The director reviews the report and either:

- **Approves**: The case is escalated to the **Assistant Commissioner**.
- **Rejects**: The case is returned to the officer with comments.

4. The **System** updates the case status and logs the action.

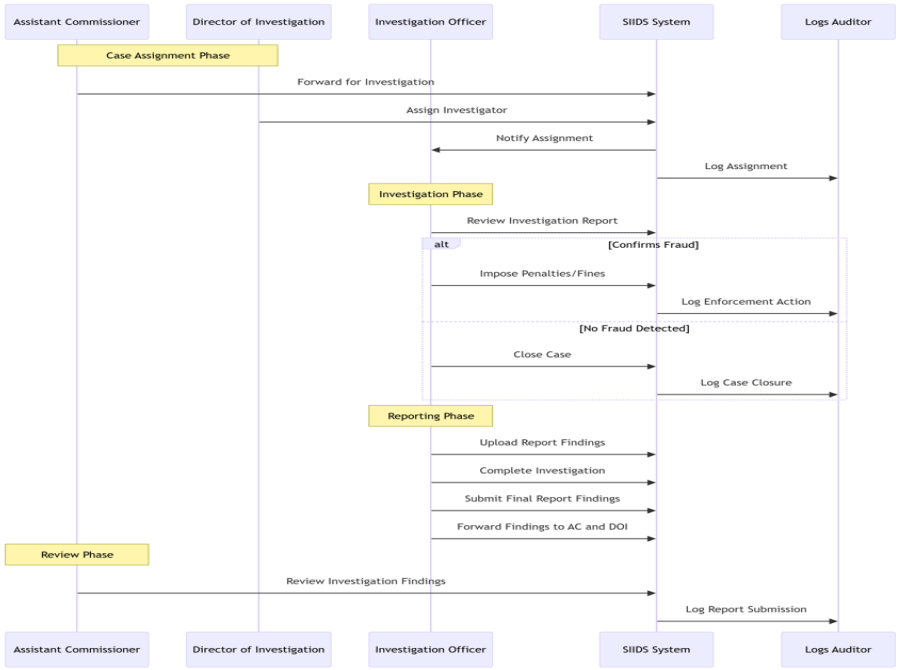


Figure 6: Investigation Workflow

Description:

- **Actors Involved**: Assistant Commissioner, Director of Investigation, Investigation Officer, System
- **Process Flow**:

1. The **Assistant Commissioner** assigns the case to the **Director of Investigation**.
2. The **Director of Investigation** selects an **Investigation Officer** and delegates the case.
3. The **System** notifies the officer and provides case details.
4. The **Investigation Officer** conducts the investigation, submits findings, and updates the case.
5. The **System** validates and stores the report, then notifies the **Assistant Commissioner**.

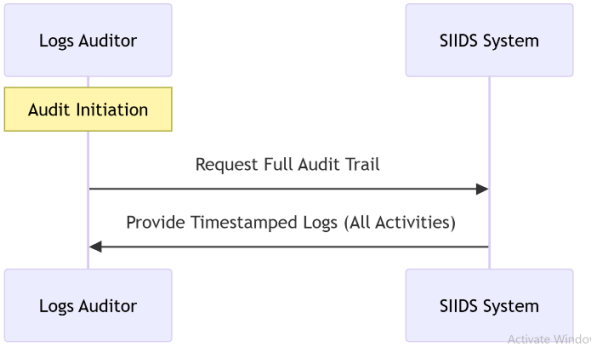


Figure 7: Audit & Compliance Monitoring

Description:

- **Actors Involved**: Logs Auditor, System
- **Process Flow**:

1. The **Logs Auditor** requests access to audit logs.
2. The **System** authenticates the auditor and retrieves logs based on filters (e.g., date, user role).
3. The auditor reviews logs for compliance (e.g., unauthorized access, case modifications).
4. The **System** generates a compliance report for export.

Activity Diagrams

The UML activity diagram serves as a crucial tool for representing system dynamics. It functions like a process flow diagram that illustrates the transitions between various system operations. Each activity represents a specific system function or operation.

Operational sequences are depicted through connecting flows, which can progress in linear paths, divide into multiple branches, or occur simultaneously. These diagrams manage different flow patterns using specialized elements, including fork and join nodes, to show how activities connect and interact.

Activity Diagram of Strategic Intelligence and Investigation Division System(SIIDS)

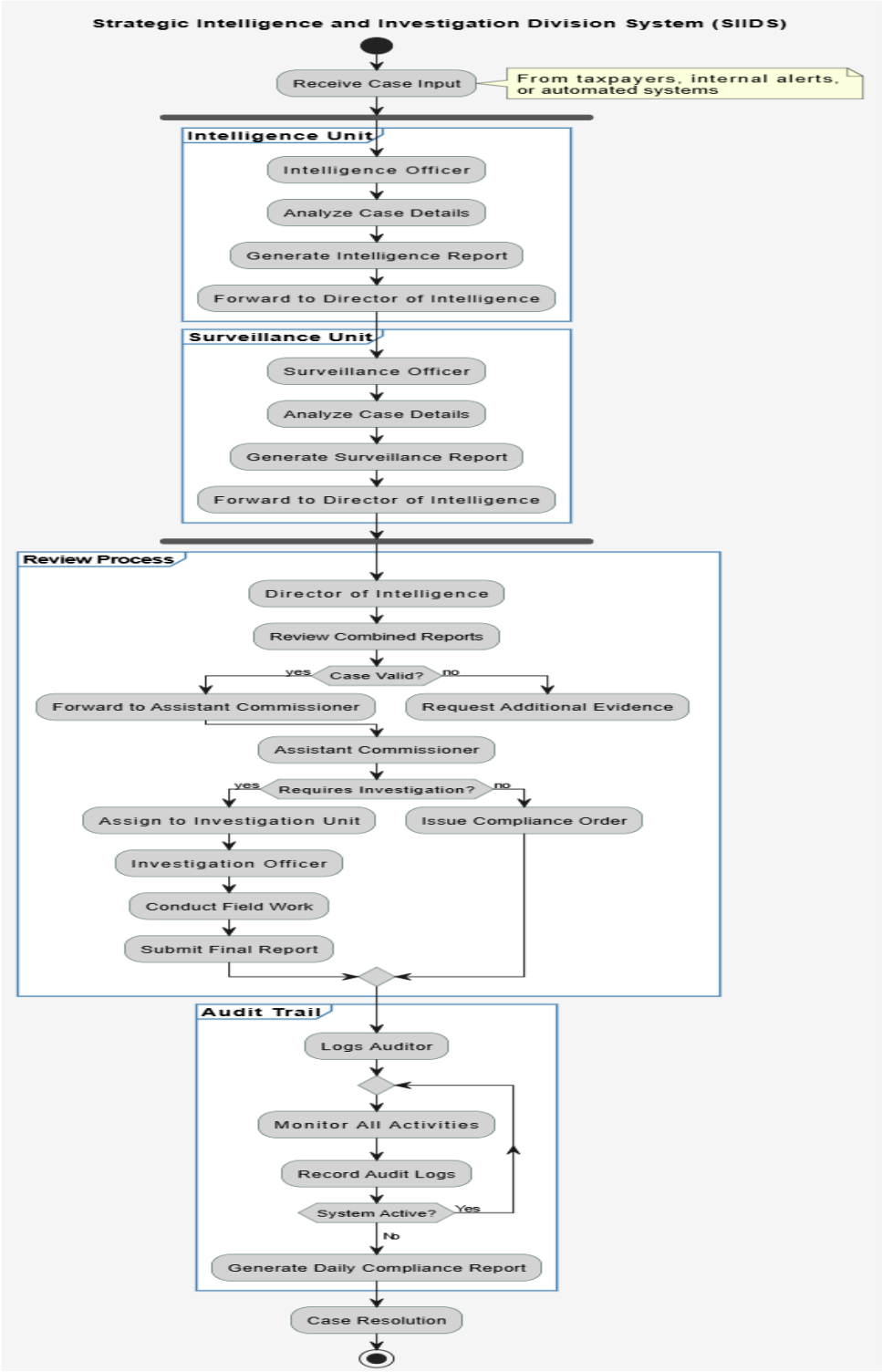


Figure 8: Activity Diagram

Database Diagrams

Visual database modeling forms an essential foundation in software development initiatives. UML Model's integration of UML database diagramming capabilities streamline the process of database visualization and management. This functionality enables developers to generate UML representations from existing database structures, maintain synchronization between diagrams and actual databases during modifications, and architect new database schemas including tables and their interconnections. (Hoffman, 2018)

Terms used and their definition	**Symbol**
An entity refers to an object or concept for which information needs to be stored.	
Cardinality defines the number of instances of one entity that can be linked to a single instance of another entity. It represents the frequency of a relationship between entities.	
Connecting line: is a solid line used to establish a one-to-one connection between attributes, illustrating entity relationships within a diagram..	
An attribute is a data element that defines the properties of a field in a database or a sequence of characters in a display.	
Relationships describe the associations between multiple entities..	

Table 21: Database diagram symbols meaning

Database Diagram of Strategic Intelligence and Investigation Division System(SIIDS)

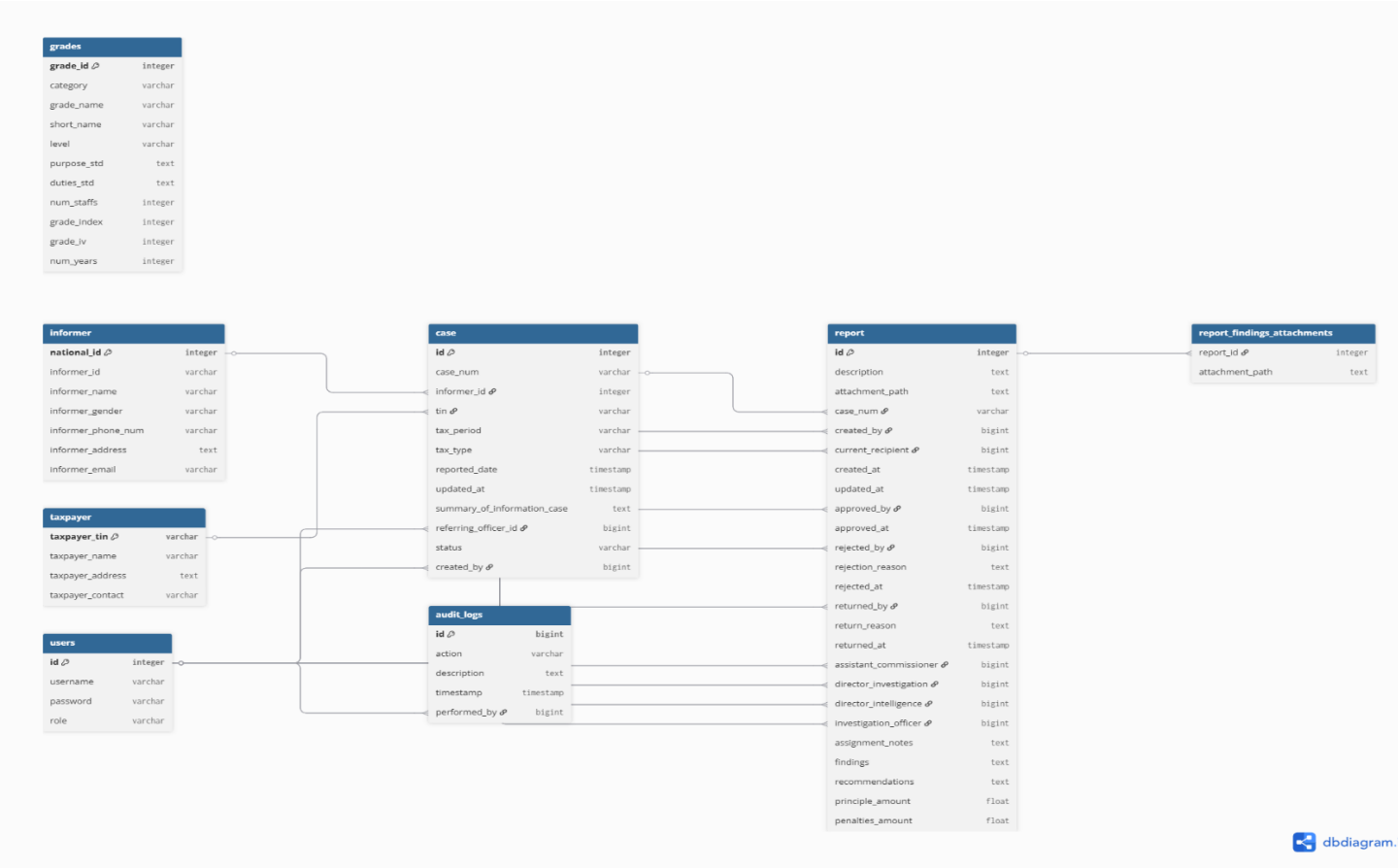


Figure 9: Database Diagram

Data Dictionary

The Data Dictionary provides detailed descriptions of the data elements used in the new system, including the tables, fields, data types, and constraints.

Case Table

Field Name	**Data TYPE**	**Constraints**	**Description**
Id	INT	AUTO_INCREMENT	Unique case identifier

CaseNum	VARCHAR(20)	UNIQUE, NOT NULL	Human-readable case number
informer_id	INT	FOREIGN KEY → informer.national_id	Case informer reference
tin	VARCHAR(20)	FOREIGN KEY → TaxPayer.TIN	Taxpayer Identification Number
Tax_Period	VARCHAR(20)		Tax period under investigation
Tax_Type	VARCHAR(20)		Type of tax (VAT, Income Tax, etc.)
reported_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Initial report date
LastUpdated	DATETIME	YYYY-MM-DD HH:MM:SS	Latest modification date
summary_of_information_case	TEXT		Case summary and details
referring_officer_id	VARCHAR(20)	FOREIGN KEY → users.id	Officer who reported the case
status	WorkFlow	FOREIGN KEY → WorkFlow	Current case status
created_by	INT	FOREIGN KEY → users.id	User who created the case

Table 20: Case Table Data Diction

Report Table

Field	Type	Format/Value	Description
Id	INT	AUTO_INCREMENT	Unique report identifier

CaseNum	INT	FK->Case.CaseNum	Linked case
description	TEXT		Report content and analysis
attachment_path	TEXT		Analysis/findings
CreatedBy	INT	FK->User.UserID	Report author
current_recipient	INT	FOREIGN KEY → users.id	Current report handler
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Report creation date
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE	Last modification
approved_by	INT	FOREIGN KEY → users.id	Approving authority
approved_at	TIMESTAMP		Approval timestamp
rejected_by			
rejection_reason	TEXT		Reason for rejection
rejected_at	TIMESTAMP		Rejection timestamp
returned_by	INT	FOREIGN KEY → users.id	User who returned report
return_reason	TEXT		Reason for return
returned_at	TIMESTAMP		Return timestamp
assistant_commissioner	INT	FOREIGN KEY → users.id	AC handling the case
director_investigation	INT	FOREIGN KEY → users.id	Investigation director
director_intelligence	INT	FOREIGN KEY → users.id	Intelligence director
investigation_officer	INT	FOREIGN KEY → users.id	Assigned investigator

findings	TEXT		Investigation findings
Recommendations	TEXT		Recommended actions
Principle	DECIMAL(15,2)		Principal tax amount owed
penalties_amount	DECIMAL(15,2)		Penalty amount imposed

Table 22: Report Table Data Dictionary

User Table

Field	Type	Format/Value	Description
id	INT	AUTO_INCREMENT	System user identifier
Username	VARCHAR(50)		User employee ID
Role	VARCHAR(50)		System role

Table 23: User Table Data Dictionary

Informer Table

Field Name	DATA TYPE	Constraints	Description
national_id	TEXT	PRIMARY KEY	National ID number
informer_id	TEXT	UNIQUE	System-generated informer ID
informer_name	TEXT	**Not Null**	Informer's full name
informer_gender	TEXT		Gender
informer_phone_num	TEXT		Contact phone number
informer_address	TEXT		Physical Address
informer_email	TEXT		Email address

Table 24: Informer Table

TaxPayer Table

Field Name	Data Type	Constraints	Description
taxpayer_tin	VARCHAR(20)	PRIMARY KEY	Taxpayer Identification Number
taxpayer_name	VARCHAR(100)	NOT NULL	Taxpayer's name/business name
taxpayer_address			Registered address
taxpayer_contact			Contact information

Table 25: Tax Payer Information

Audit Log Table

Field	Type	Format/Values	Description
ActionID	INT	AUTO_INCREMENT	Action identifier
CaseNum	INT	FK→Case.CaseNum	Linked case
ActionType	VARCHAR(20)		Type of action taken
PerformedBy	INT	FK→User.UserID	Actor executing action
ActionDate	DATETIME	YYYY-MM-DD HH:MM:SS	Timestamp of action

Table 26: Case Action Data Dictionary

System Architecture Design

A system architecture diagram visually represents a system, where key components or functions are depicted as blocks connected by lines to illustrate their relationships.

Strategic Intelligence and Investigation Division System(SIIDS)

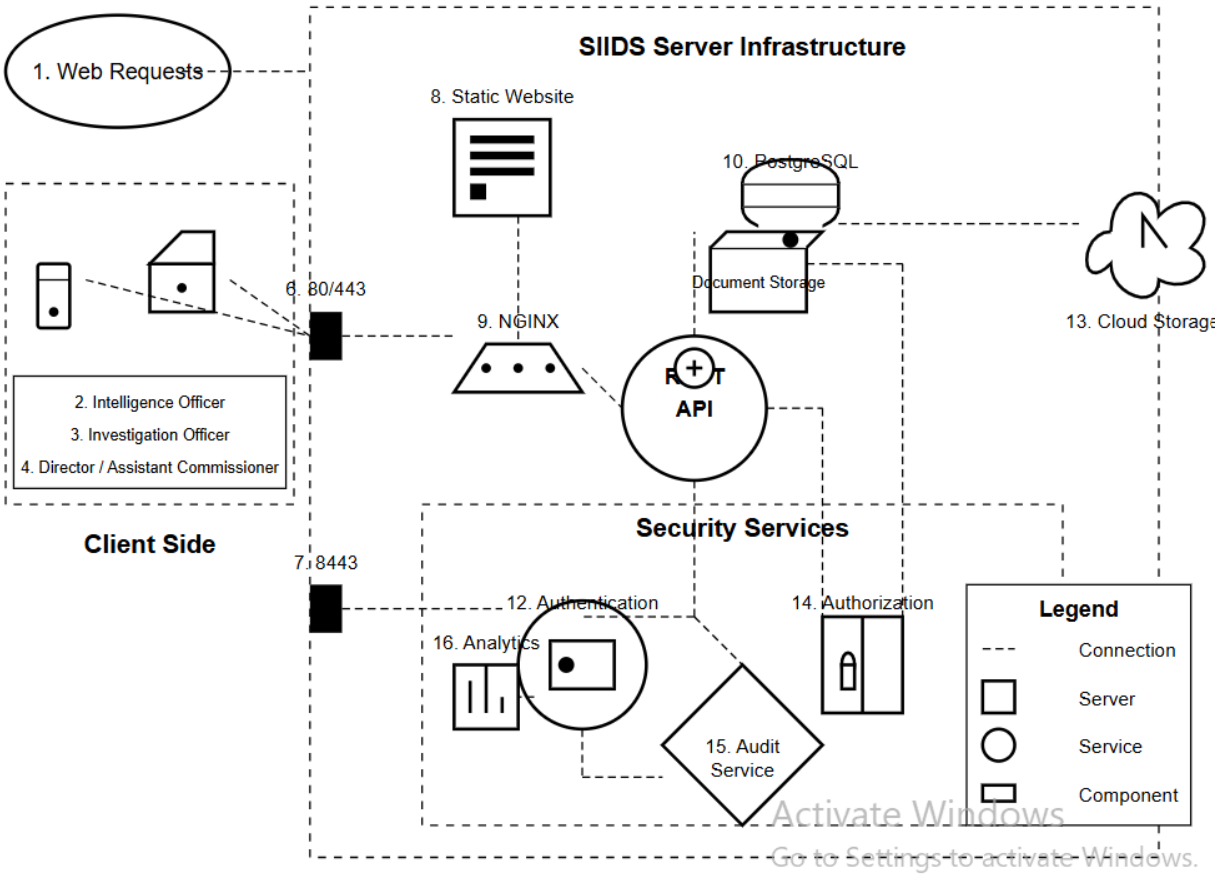


Figure 10: System architecture

CHAPTER 4

IMPLEMENTATION AND TESTING OF THE NEW SYSTEM

Introduction

This chapter outlines the implementation process of the **Strategic Intelligence and Investigation Division System** (SIIDS) and the testing strategies employed to ensure its reliability, security, and effectiveness. The implementation stage involved translating system designs into a working application, integrating front-end and back-end components, and configuring external services. Rigorous testing followed to verify that each system feature meets user expectations and technical requirements.

The chapter also describes the technologies and tools used during development, the process of building the user interface and application logic, and the execution of various software tests such as unit testing, integration testing, system testing, and user acceptance testing.

Technologies Used

Frontend

- **React.js:**
  React is a powerful JavaScript library for building user interfaces. It allows the creation of reusable, component-based designs, which improves development speed and maintainability. Its virtual DOM ensures that updates are efficiently rendered, enhancing performance, especially in data-intensive systems like SIIDS.
- **Redux:**
  Redux is used for managing application-wide state in a predictable way. In SIIDS, Redux stores and synchronizes case data, user sessions, and notifications across different parts of the application. It ensures consistency and avoids unnecessary data duplication or loss during user interactions.

- **Axios**
  Axios is used for making HTTP requests from the frontend to the backend REST APIs. It simplifies secure communication with backend services and handles error responses and token-based authentication headers.
- **Bootstrap/Tailwind CSS**
  These frameworks help quickly style and structure the user interface to be responsive and clean, ensuring accessibility across different screen sizes and devices.

Backend

The back-end is the core processing engine of the system. It handles authentication, business logic, database interactions, and external integrations.

- **Spring Boot (Java)**
  Spring Boot is a production-grade framework for building secure, scalable, and robust Java applications. It simplifies the configuration of enterprise-grade APIs and enforces best practices such as separation of concerns through layered architecture (controllers, services, and repositories). Spring Boot was ideal for this project due to its:
  - Built-in support for REST APIs
  - Security integration (e.g., Spring Security with JWT)
  - Compatibility with PostgreSQL
  - Developer productivity and community support
- **Spring Security with JWT**
  Used to manage user authentication and authorization. JWT (JSON Web Tokens) allows for stateless, secure session handling, ensuring only authorized users can access specific features based on roles.
- **PostgreSQL**
  A reliable and powerful open-source relational database used to store structured case information, user profiles, reports, and logs. It supports foreign keys, complex queries, and secure access control.

- **RESTful API Architecture**

  Backend endpoints are built using REST principles, enabling modular, maintainable, and scalable client-server interactions. This ensures that the frontend and external systems can securely access and exchange data with the backend.

Presentation of the New System

Login Page:

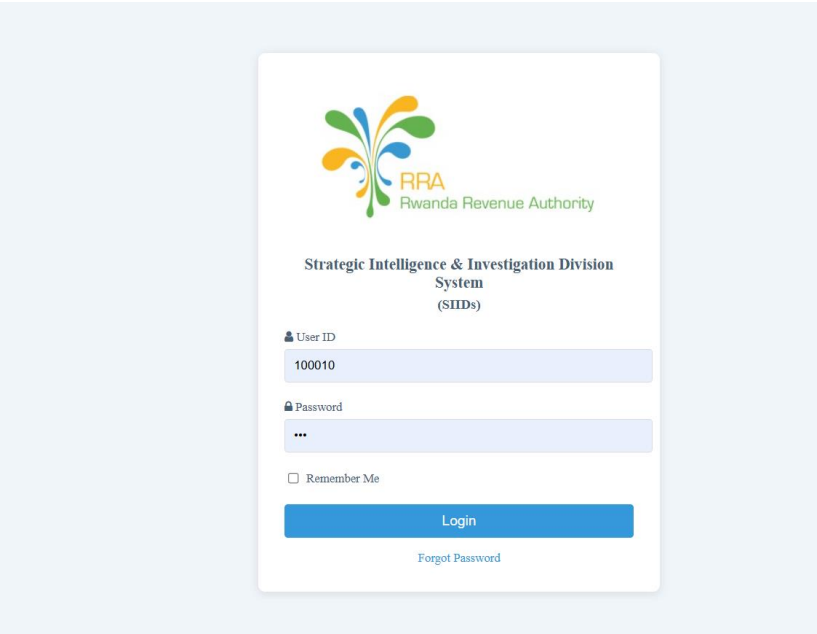


Figure 11: Login Page

Intelligence Officer Page:

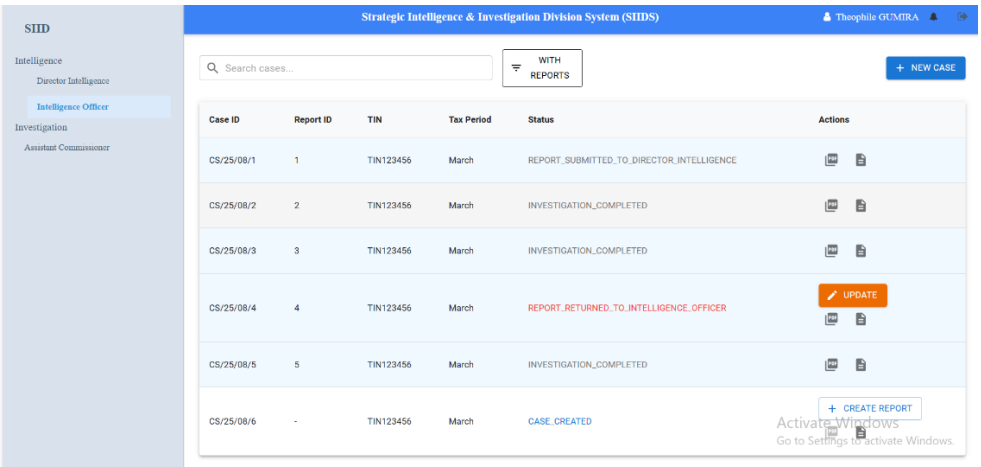


Figure 12: Intelligence Officer Page

New Case Created

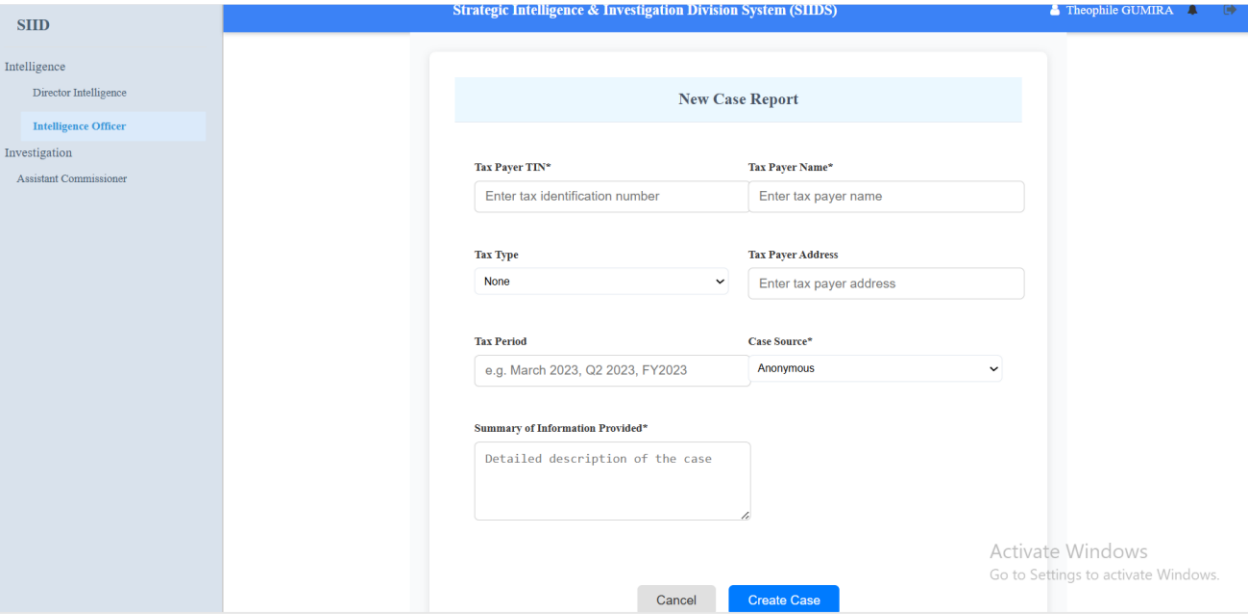


Figure 13: New Case Creation page

Director of Intelligence page:

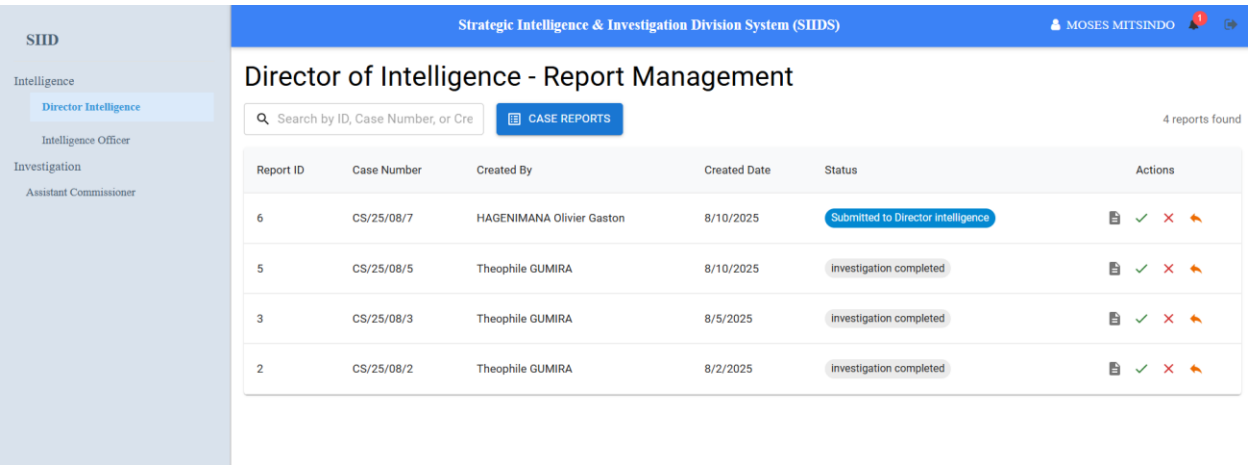


Figure 14: Director of Intelligence page

Assistant Commissioner Page:

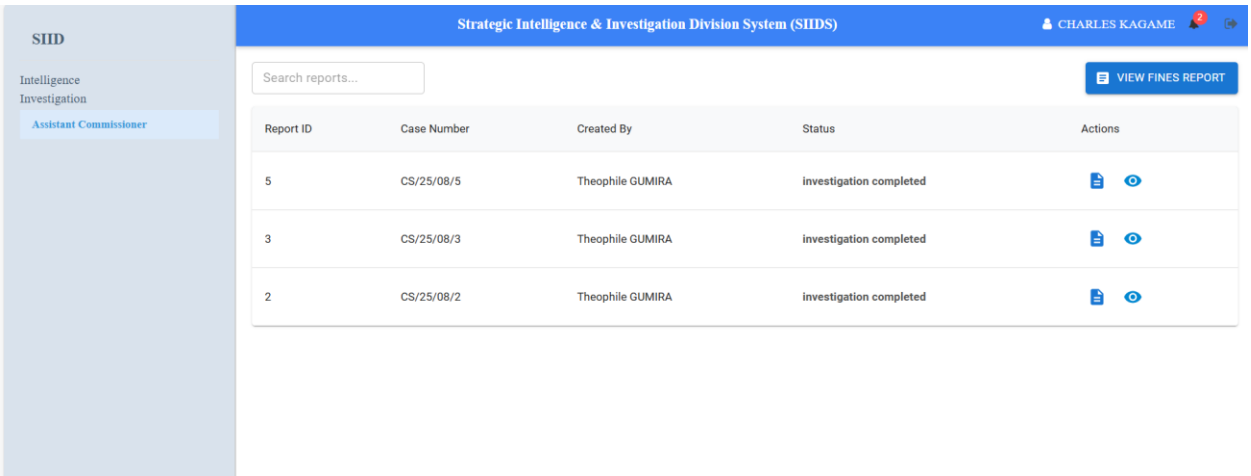


Figure 15: Assistant Commissioner Page

Director of Investigation Page:

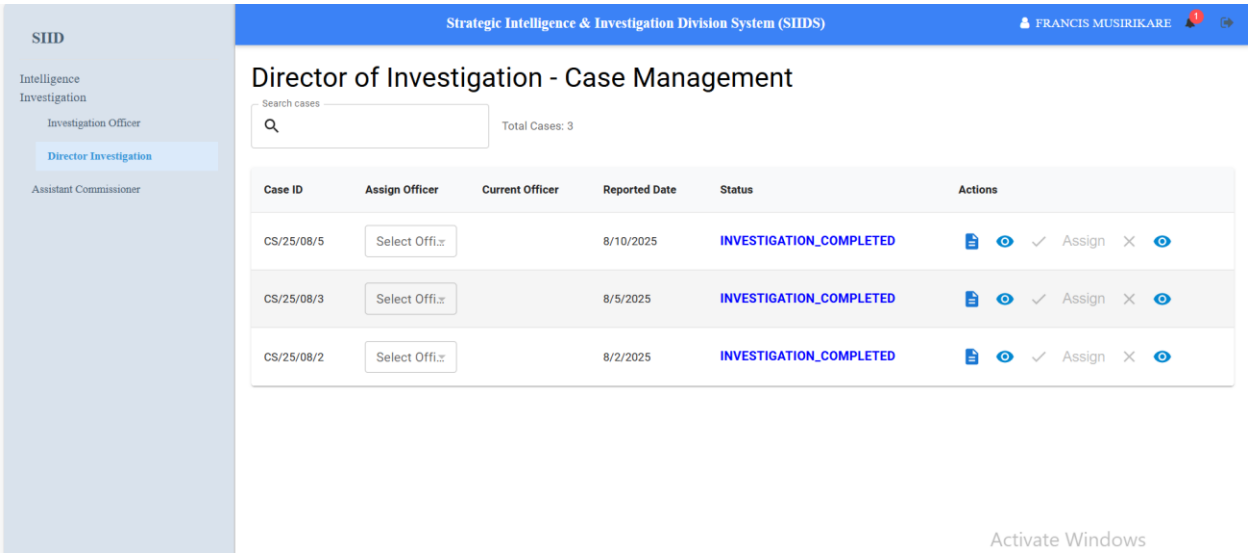


Figure 16: Director of Investigation page

Investigation Officer Page:

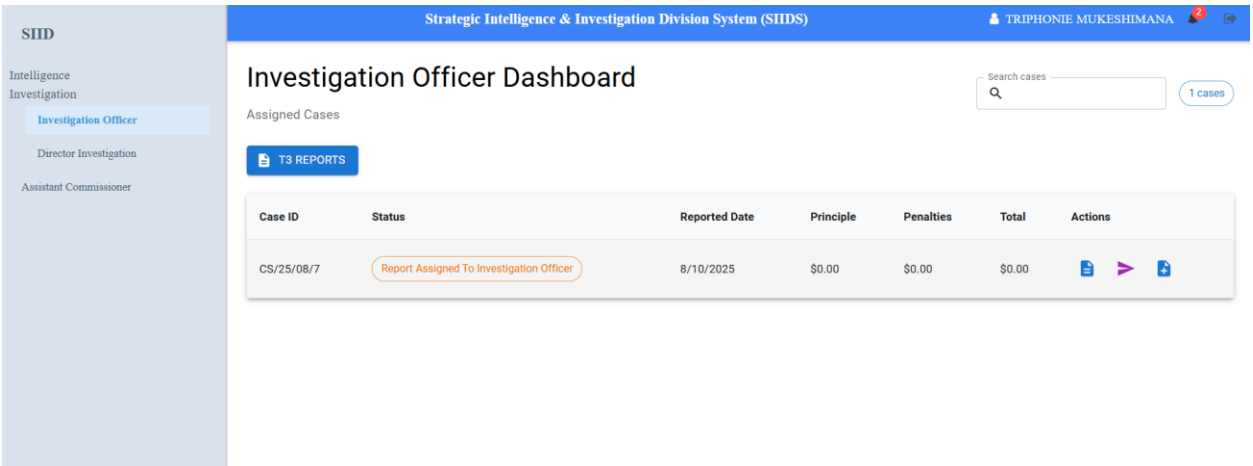


Figure 17: Investigation Officer Page

Audit Log Page:

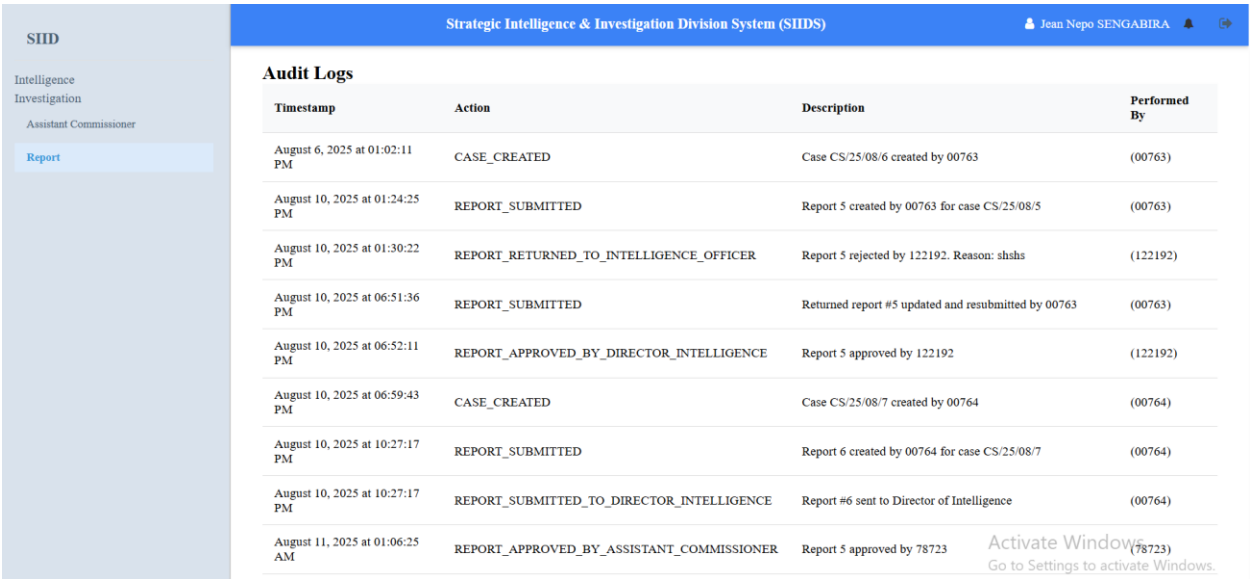


Figure 18: Audit logs Page

Software Testing

Software testing plays a crucial role in the design process, helping to determine whether the software effectively performs its intended functions.

When conducting software testing, the following aspects should be considered:

- Does the application comply with the standards set during its design and development?

- Is the application functioning correctly?
- Can the application be deployed in the intended manner while meeting stakeholder requirements?

Below are some key types of software testing:

Unit Testing

Unit testing is a method used to verify the proper functioning of individual components within a software system. It involves evaluating specific pieces of source code, program modules, control data, usage processes, and operating procedures to ensure they meet their specifications and are free from logical errors. Each unit is tested independently, making unit testing a highly effective tool for detecting errors. Every newly written section of code undergoes unit testing before being integrated into the system.

Integration Testing

This phase of testing involves combining multiple software components and verifying their correct interaction. It ensures that the various modules of the software integrate as intended. As more tests are conducted, both hardware and software components are assembled and evaluated until the complete system is tested. Each module is individually examined to ensure that all software components meet the required functional and technical specifications before final integration.

Validation Testing

The final phase of testing focuses on validating the software within its external environment. This step ensures that the application meets the initial requirements outlined during the design phase. Validation testing confirms that the application's configuration aligns with the specified needs. For example, the system should enforce a 10-digit phone number format, and email addresses must end with "@rra.gov.rw". The application underwent thorough validation testing, confirming that its operations align with the defined functional requirements.

User Acceptance Testing

User Acceptance Testing (UAT) was conducted to ensure the platform meets the expectations and needs of its intended users, including farmers and landowners. This phase focused on validating the platform's usability, functionality, and overall performance under real-world conditions.

Key findings from UAT include:

- **Response Time**: The platform demonstrated an average response time of 1.8 seconds when loading map functionality and profile pages, ensuring a smooth user experience.
- **User Feedback**: Feedback revealed that 90% found the platform intuitive and easy to navigate, emphasizing its user-centered design.
- **Task Completion Rate**: 85% of users successfully completed critical tasks such as creating profiles, listing land, and explore available land without requiring external assistance.
- **Error Reduction**: Iterative testing during UAT led to a 93% decrease in error reports for core functionalities, such as automated notifications and lease agreement processing.

These results validated that the platform aligns with user expectations, offering a seamless and efficient experience for both farmers and landowners. The insights gained from this phase informed further refinements to enhance usability and ensure the platform's performance.

Hardware and Software Requirements

These requirements apply to RRA staff and users (Intelligence Officers, Investigators, and Administrators) who access the system through web browsers.

Client Side Requirements

Client-Side Software Requirements:

SOFTWARE	MINIMUM VERSION	PURPOSE

WEB BROWSER	Chrome 90+, Firefox 88+, Edge 90+	Accessing the SIIDS platform securely
OPERATING SYSTEM	Windows 10 / Ubuntu 20.04 / macOS 11+	System compatibility and performance
PDF READER (OPTIONAL)	Adobe Reader / Web PDF Viewer	Viewing and exporting generated reports
OFFICE SUITE (OPTIONAL)	MS Excel / LibreOffice	Opening exported reports (CSV, Excel format)
INTERNET CONNECTIVITY	Stable connection (≥10 Mbps)	Required for secure communication with backend

Table 27: Client-Side Software Requirements

Client-Side Hardware Requirements:

Hardware	***Minimum Specification***
Processor	Dual-core 2.0 GHz or higher
RAM	4 GB minimum (8 GB recommended)
Storage	At least 10 GB free disk space
Display	13” screen (minimum 1366×768 resolution)
Network Interface	Ethernet or Wi-Fi adapter

Table 28: Client-Side Hardware Requirements

Server Side Requirements

Server-Side Software Requirements:

Software	Purpose
Ubuntu Server 20.04 LTS / CentOS / Windows Server	Operating System for backend deployment

Java JDK 17+	Running the Spring Boot backend application
PostgreSQL 14+	Database engine for storing system data
NGINX / Apache HTTP Server	Serving the frontend and managing routing
Docker & Docker Compose (Optional)	Containerized deployment and orchestration
SSL Certificate	Secure HTTPS communication
Git	Source code version control and updates

Table 29: Server-Software Requirements

Server-Side Hardware Requirements:

Hardware	***Minimum Specification***
Processor	Quad-core 2.5 GHz or higher
RAM	16 GB (32 GB recommended for peak performance)
Storage	256 GB SSD or higher
Bandwidth	100 Mbps or more (symmetric upload/download)
Redundancy	RAID setup and backup systems recommended

Table 30: Server-Side Hardware Requirements

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

Conclusion

The development of the Strategic Intelligence and Investigation Division System (SIIDS) represents a critical step toward the modernization and digital transformation of the Rwanda Revenue Authority's (RRA) intelligence and investigation operations. This project was driven by the need to address inefficiencies, delays, and security concerns inherent in the current manual and semi-automated systems. Through detailed analysis, design, implementation, and testing, SIIDS was developed to provide a centralized, intelligent, and secure platform that supports real-time case management, fraud detection, and decision-making.

The project began with an in-depth review of the existing system environment, identifying challenges such as fragmented data sources, delayed case processing, and lack of transparency in investigation workflows. A layered system architecture was then proposed, incorporating modern technologies including React.js for the front-end, Spring Boot for the backend, and PostgreSQL for data management. Security features like JWT-based authentication and role-based access control were also integrated to protect sensitive information.

One of the key strengths of the project was the use of Agile methodology, allowing iterative development and regular feedback from stakeholders. Diagrams such as use case, class, activity, and sequence diagrams helped translate user requirements into practical design blueprints. Functional and non-functional requirements were clearly defined, ensuring that the system was not only technically sound but also aligned with operational realities.

Testing played a vital role in verifying the system's performance, reliability, and security. Unit, integration, system, and user acceptance tests were conducted to validate each component. The results confirmed that SIIDS successfully met the identified needs, such as automating case

tracking, generating real-time reports, integrating with national databases, and providing secure, role-based access.

In conclusion, SIIDS stands as a scalable, secure, and efficient solution that can significantly enhance the effectiveness of tax fraud investigation and compliance enforcement at RRA. It supports the institution's broader goals of good governance, transparency, and digital innovation. With future enhancements such as machine learning for predictive fraud detection and integration with financial intelligence units, SIIDS has the potential to serve as a model for other government agencies pursuing digital transformation in public service delivery.

Recommendations

To ensure the long-term success and sustainability of the Strategic Intelligence and Investigation Division System (SIIDS), the Rwanda Revenue Authority (RRA) should prioritize structured training and capacity building. All system users including Intelligence Officers, Investigators, and Administrators should receive role-specific training to ensure they understand how to navigate and utilize the platform efficiently. This will reduce user errors, increase adoption, and promote consistency across departments.

RRA should also establish a dedicated technical support and maintenance team. This team would be responsible for regular system updates, patching security vulnerabilities, and ensuring that the system evolves in response to changing organizational needs. A well-maintained system will reduce downtime, enhance user trust, and improve overall performance.

To strengthen investigative capabilities, RRA is encouraged to expand SIIDS integration with other government and regulatory bodies. Future integrations could include the National Public Prosecution Authority (NPPA), Rwanda National Police (RNP), and financial institutions. These partnerships would enrich the data available for fraud detection and improve inter-agency collaboration.

Feedback mechanisms should be built into the system to allow users to report issues, suggest improvements, and share experiences. Regular review sessions can then be used to evaluate

feedback and identify areas for enhancement. This approach ensures that the system remains user-centered and adaptive to real-world needs.

It is also recommended that RRA explore the use of artificial intelligence and predictive analytics to further strengthen fraud detection. These tools could help identify risky taxpayer behavior before it escalates into full-scale fraud cases, allowing for proactive investigations and interventions.

Finally, before nationwide rollout, the system should undergo pilot testing in selected departments or regions. This would allow developers and stakeholders to observe real-world usage, collect feedback, and refine the platform before full deployment.

By implementing these recommendations, RRA can maximize the impact of SIIDS, ensure long-term usability, and contribute to a more transparent, efficient, and digitally empowered tax investigation environment.

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APPENDICES

Data Collection

Adventist University of Central Africa
P.O. Box 2461 Kigali, Rwanda | www.auca.ac.rw | info@auca.ac.rw

Faculty of Information Technology

January 27, 2025

TO: RWANDA REVENUE AUTHORITY (RRA)

Dear Sir/Madam,

Re: Request for Data Collection

The bearer, **NTABANA Nelson** is a student of Adventist University of Central Africa in the Faculty of Information Technology, major in Software Engineering Department and he is working on a dissertation entitled: **"STRATEGIC INTELLIGENCE AND INVESTIGATION DIVISION SYSTEM (SIIDS)"**

The purpose of this letter is to request for your permission to allow him to collect data in your organization.

Any assistance given to him will be highly appreciated.

Yours sincerely:

Mr. ISHIMWE Prince
HOD||Software Engineering
Tel: +250 782 974 477
Email: prince.ishimwe@auca.ac.rw

Approval letter

Application Unit
Rwanda Revenue Authority
Kigali-Rwanda
11th/ November/ 2024

To:
Adventist University of Central Africa
Kigali-Rwanda

To Whom It May Concern

RE: Approval for the Strategic Intelligence and Investigation Division System (SIIDS)

This letter is to formally approve the submission of Nelson NTABANA on the Strategic Intelligence and Investigation Division System (SIIDS) project proposal. The project is designed to address intelligence, investigation, and surveillance activities, and aligns with organizational needs and objectives.

We are confident that the proposed system will enable SI&ID staff to log in and perform their daily activities efficiently. The outlined features, including the management all information received by Intelligence and Surveillance Units, case tracking, enforcement step access, assessed tax capturing, and integration with RRA software and external stakeholders' systems, demonstrate its relevance and potential impact.

The system's design emphasizes user-friendly performance, 24/7 availability, high-speed responsiveness across web browsers, and robust security measures, such as disaster recovery plans, regular backups, and alerts for unauthorized access attempts. These features will significantly minimize risks associated with manual operations, ensuring a more efficient and secure workflow.

We are eager to see the development of SIIDS and its potential to enhance the organization's strategic intelligence and investigative capabilities.

Yours sincerely,

BISENGIMANA IVAN

RWANDA REVENUE AUTHORITY
P.O. BOX 3987 KIGALI

curriculum vitae (CV)

NTABANA NELSON

SOFTWARE DEVELOPER

CONTACT

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KN 217st

linkedin.com/in/ntabana-nelson-a81702295

PROFILE

Detail-oriented and innovative Software Engineer with hands-on experience in developing web-based applications. Skilled in Java, Spring Boot, and React, with a strong focus on web systems and enterprise applications. Adept at working in Agile environments, collaborating with cross-functional teams, and delivering high-quality software solutions. Passionate about continuous learning and professional growth in software development.

SKILLS

- Strategic Planning
- Problem Solving
- Crisis Management
- Creative Thinking
- Data Analysis
- Adaptability to Change

WORK EXPERIENCE

Rwanda Revenue Authority (July 2024 - Present)

Software Engineer

- Developed a web-based tax management system using Spring Boot and React, improving processing efficiency by 30%.
- Implemented JWT authentication to enhance security for user access and session management, significantly reducing unauthorized access incidents.
- Collaborated with Agile teams, participating in sprints and code reviews to improve software quality and team performance.
- Optimized database queries, reducing response times and improving system performance.

LANGUAGES

- English (Fluent)
- Kinyarwanda (Native)
- French (Intermediate)

EDUCATION

Adventist University of Central Africa (AUCA) 2021-2025

Bachelor of Science in Information Technology

(Major in Software Engineering)

St Mary's College Kiabi 2014-2018

Ordinary Level & Advanced Level

TECHNICAL SKILLS

- Java, JavaScript, C#
- Spring Boot, ASP.NET Core, Hibernate, React, Redux, Material-UI
- PostgreSQL, SQL Server
- REST APIs, JWT Authentication, Git, Agile Methodologies

PROJECT

Adventist University of Central Africa (AUCA)

Bus Transport Management System

- Technologies: C# (.NET Core), ASP.NET Core, SQL Server
- Developed to implement features for real-time bus tracking, dynamic pricing algorithms, and automated passenger boarding management.
- It managed and tracked bus transportation operations.

CERTIFICATION

- Cisco Networking Essentials – Cisco Networking Academy
- Linux Essentials – Cisco Networking Academy
- Bridge Program – Carnegie Mellon University (CMU)
- NetOps 1

Rwanda Revenue Authority Nov 2024 - Jan 2025

Meeting Room Management System

- Built a role-based room booking system with secure authentication and access control using Spring Boot and React.
- Utilized JWT authentication for streamlined and secure user login and role management.

Jan 2025 – April 2025

Training Management System

(Rwanda Revenue Authority)

- Developed a comprehensive training management system tailored for RRA employees using Spring Boot, React, and PostgreSQL.
- Ensured seamless data handling and intuitive user experience through optimized UI/UX design.

School Management System

- Developed a robust system to streamline school operations, including student enrollment, attendance tracking, grade management, and staff scheduling.
- Utilized ASP.NET Core scaffolding to rapidly generate CRUD interfaces for core entities like students, teachers, subjects, and classes, then customized logic and UI for specific business needs.

REFERENCES

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